

Logistic Biclustering Models for Protein Network Inference

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Abstract

Protein interaction information is fundamental to understand the cellular processes. Due to that, much is being done to automatically infer protein networks from all types of biological data. In this work we propose a model based on logistic biclustering, to analyze biological data in binary format, and combine that with a plaid model analysis of microarray expression data to predict protein interactions. We compare the results with a supervised model applied to the same data sets, and the experiments show that the proposed method achieves a performance very close to the one of a supervised approach trained with up to 25% of knowledge of the target network.

1. Introduction

Making inference about protein network has been an issue of great relevance, since protein interactions are related to their biological functions. Genomic data may be generated in several ways. Since each one of them may represent new relevant genomic information, techniques for integrating them into a single method are needed in order to make inference about protein networks. Several strategies have been proposed in the literature, some of them are based on the impact of domain interactions on the network likelihood [10], others use parsimony to try to explain the protein network [6], and still others estimate structure or sequence conservation [2, 5]. Due to the noise and the incompleteness of virtually all kinds of biological data, a trend of integration has been established in the last years to profit from different sources [1, 10, 12].

Data integration is not only done through different types of data regarding the same organism, but also by using the same kind of data associated to different organisms. An example of the first is the work of Yamanishi and colleagues [20], who proposed a supervised approach to make inference for protein networks taking into account information from four types of genomic data sets for the same organism. The latter is applied, for instance, by Reiss and colleagues [15] who work with integrated

biclustering to make inference of protein networks using only expression data, but considering four different organisms.

In Yamanishi's approach, they used a kernel representation for each type of genomic data, and then integrated the kernel matrices using a simple arithmetic mean. That integrated kernel was used to perform a supervised training through kernel canonical correlation analysis (KCCA), using knowledge of a subset of the gold standard protein network; that is, it takes information of part of the protein network itself for which they wanted to make inference.

A useful technique to discover associations among sets of genes is the biclustering of gene expression microarray data, which provides a better understanding of the biological structure behind them [11]. Gene expression data is usually presented in a matrix form, with r rows representing the genes, c columns representing the conditions, and Y_{ij} the elements of the matrix, representing a measure of expression level of gene i under condition j ($i=1, \dots, r$ and $j=1, \dots, c$).

Microarray expression measures usually are real values, and approaches such as the *Plaid Model* [9, 19] analyze them using tools from *Two-Way Analysis of Variance* [17]. However, there exists a variety of genomic data for which the only recorded information about gene i under condition j is a binary variable representing the presence or absence of a given characteristic. Examples of these data can be found, for instance, in phylogenetic profiles of proteins [13] or converting nominal values into binary data. The analysis of such types of data using models designed to handle real values, as commonly done in searching methods for clustering, like *K-means* [14] or the *Plaid Model*, may be less effective or even not possible. Additionally, the binary data is modeled as real values between 0 and 1, which do not properly represent them.

We propose a model, *Lbic Model*, for analyzing binary data that uses an approach similar to a *Two-Way Logistic Analysis* in order to find biclusters. A bicluster is a sub-matrix of a genomic data matrix such that the genes represented by its lines are more homogeneous under the subset of conditions represented by its columns than under the conditions

in general. We show that the proposed method has a much better performance than the Plaid Model when applied to the same data set. Additionally, the Lbic model shows to be a better tool to predict protein networks from phylogenetic data when compared to supervised method KCCA, even without prior knowledge.

Further we propose a way to combine results from two different kinds of genomic data using the Lbic model applied to *Phylogenetic profiles* and the Plaid model applied to *Expression data*. We show that this combination has a performance very similar to the supervised method KCCA when trained with up to 25% of the gold standard network, using the same source data.

2. Plaid Model

Let $A = (Y_{ij})$, $i = 1, \dots, n$ and $j = 1, \dots, c$, be a matrix of the expression data, where the rows represent the genes, the columns represent the experiments, and Y_{ij} represents the expression of gene i under experiment j . Let $\theta_{ijk} = \mu_k + \alpha_{ik} + \beta_{jk}$ be the effect of bicluster k for gene i and experiment j , with $k = 1, \dots, K$; $i = 1, \dots, r$, and $j = 1, \dots, c$, where μ_k is the mean effect of bicluster k , α_{ik} is the effect of gene i on bicluster k , and β_{jk} is the effect of condition j on bicluster k . Then the expression of gene i in experiment j is given as the sum of effects of all biclusters which contain gene i under experiment j , as follows.

$$Y_{ij} = \sum_{k=1}^K \theta_{ijk} \rho_{ik} \kappa_{jk} \quad (1)$$

$$\rho_{ik} = \begin{cases} 1 & \text{if the gene } i \text{ is in the bicluster } k \\ 0 & \text{otherwise} \end{cases}$$

$$\kappa_{jk} = \begin{cases} 1 & \text{if the condition } j \text{ is in the bicluster } k \\ 0 & \text{otherwise} \end{cases}$$

This algorithm splits the original matrix into a set of biclusters found through an evaluation of the sum of the square of error, SSE, given by

$$\sum_{k=1}^K (Y_{ijk} - \theta_{ijk} \rho_{ik} \kappa_{jk})^2 / 2\sigma^2$$

3. Lbic Model

As the Plaid model, the Lbic model consists of several layers trying to explain the association between some genes and conditions. The conditions depend on the data set, for expression data, those are

experiments, and for phylogenetic data, those are organisms. Let $A = (Y_{ij})$, $i = 1, \dots, n$ and $j = 1, \dots, c$, a matrix of the phylogenetic profiles, where the rows represent the genes, the columns represent the organisms, and Y_{ij} represents the presence ($Y_{ij} = 1$) or absence ($Y_{ij} = 0$) of an orthologous gene i in organism j . First we intend to model the effect of the bicluster k as $\theta_{ijk} = \mu_k + \alpha_{ik} + \beta_{jk}$, $k = 1, \dots, K$; $i = 1, \dots, r$ and $j = 1, \dots, c$, for gene i and organism j , where μ_k is the mean effect of the bicluster k , α_{ik} is the effect of gene i on bicluster k , and β_{jk} is the effect of organism j on bicluster k . Second we define $P(Y_{ij} = 1 | \theta_{ij}) = \pi(\theta_{ij})$ be the conditional probability of $Y_{ij} = 1$, given all bicluster effects in which i and j appear together. Further, we model the *logit function* [3,7] $g(\pi(\theta_{ij})) = \log[\pi(\theta_{ij}) / (1 - \pi(\theta_{ij}))]$ as

$$g(\pi(\theta_{ij})) = \sum_{k=1}^K \theta_{ijk} \rho_{ik} \kappa_{jk}, \quad (2)$$

where ρ and κ are as given in (1), and $g(\pi(\theta_{ij}))$ represents the sum of all biclusters effects for gene i and organism j .

Our algorithm also splits the original matrix into a set of biclusters found through an evaluation of the sum of the square of error (SSE), given by

$$SSE = \sum_{ij} \left(Y_{ij} - \frac{\exp\left[\sum_{k=1}^K (\mu_k + \alpha_{ik} + \beta_{jk}) \rho_{ik} \kappa_{jk}\right]}{1 + \exp\left[\sum_{k=1}^K (\mu_k + \alpha_{ik} + \beta_{jk}) \rho_{ik} \kappa_{jk}\right]} \right)^2$$

This algorithm searches for one bicluster at a time. As in the Plaid model, the layers in the Lbic model are added until a pre-specified number of biclusters is reached or no more significant layers can be found, or yet the model does not converge. Since the number of pairs of genes that in fact interact is too small as compared to the total of potential pairs of genes, and it is known that only a small group of genes have some kind of relationship, we are interested in biclusters as small as possible.

The outline of the Lbic algorithm is given below. (The presentation format is the same one used for Algorithm 1 in Tuner et al. [19].)

Algorithm Lbic

1. Compute starting values for ρ_i^0 using a sample of k-means cluster, $k=4$.
2. Compute starting values for κ_j^0 using the biggest k-means cluster, $k=2$.
3. Set $s=1$.
4. Run the logistic model using the submatrix of A indicated by ρ_i^{s-1} and κ_j^{s-1} .
5. Update layer effects using the estimated parameters in step 4.
6. Update cluster membership parameters.

$$\rho_{ik}^s = \begin{cases} 1 & \text{if } \sum_{j: \kappa_{jk}=1} Y_{ij} - \frac{\exp[\sum_{l=1}^{K_i} (\mu_k + \alpha_{ik} + \beta_{jk}) \kappa_{jk}^{s-1}]}{1 + \exp[\sum_{l=1}^{K_i} (\mu_k + \alpha_{ik} + \beta_{jk}) \kappa_{jk}^{s-1}]} < \\ & (1 - \gamma) \sum_{j: \kappa_{jk}=1} (Y_{ij} - 0.5 * \kappa_{jk}^{s-1}) \\ 0 & \text{otherwise} \end{cases}$$

and

$$\kappa_{jk}^s = \begin{cases} 1 & \text{if } \sum_{i: \rho_{ik}=1} Y_{ij} - \frac{\exp[\sum_{l=1}^{K_i} (\mu_k + \alpha_{ik} + \beta_{jk}) \rho_{ik}^{s-1}]}{1 + \exp[\sum_{l=1}^{K_i} (\mu_k + \alpha_{ik} + \beta_{jk}) \rho_{ik}^{s-1}]} < \\ & (1 - \gamma) \sum_{j: \kappa_{jk}=1} (Y_{ij} - 0.5 * \rho_{ik}^{s-1}) \\ 0 & \text{otherwise} \end{cases}$$

7. Repeat steps 4, 5 and 6 for $s = 2, \dots, S$.
8. Calculate the its mean square error (MSE) of layer, indicated by ρ_i^S and κ_j^S .
9. Permute the rows and the columns of A and follow steps 1 through 8.
10. Repeat step 9 K times.
11. Accept this layer as a bicluster if its MSE is smaller than the MSE of each one of the permuted runs; otherwise stop.

For phylogenetic data we expect that the bigger the number of organisms sharing a given pair of genes, the higher the chances that those genes interact. Hence, the layer that we look for has few genes and lots of conditions. The algorithm has an iterative process using the information of the previous state of the layer. To initialize the algorithm we present starting values for ρ and κ ; in order to get an initial layer; those values are obtained by k-means clustering [8]. We first apply k-means, $k=4$, to cluster the genes, and randomly choose one of the four clusters generated. Then we independently apply k-means, $k = 2$, to cluster the conditions in the original matrix, and take the biggest cluster. The starting layer is defined by the genes chosen in the first clustering and the conditions in the second one. This initial layer is modeled by the logit function given in Expression (2), and the new values for ρ and κ are found to satisfy:

$$\rho_{ik} = \begin{cases} 1 & \text{if } \sum_{j: \kappa_{jk}=1} Y_{ij} - \frac{\exp[\sum_{l=1}^K (\mu_k + \alpha_{ik} + \beta_{jk}) \kappa_{jk}]}{1 + \exp[\sum_{l=1}^K (\mu_k + \alpha_{ik} + \beta_{jk}) \kappa_{jk}]} < \\ & (1 - \gamma) \sum_{j: \kappa_{jk}=1} (Y_{ij} - 0.5 * \kappa_{jk}) \quad \text{and} \quad \tilde{\rho}_{ik} = 1 \\ 0 & \text{otherwise} \end{cases}$$

and

$$\kappa_{jk} = \begin{cases} 1 & \text{if } \sum_{i: \rho_{ik}=1} Y_{ij} - \frac{\exp[\sum_{l=1}^K (\mu_k + \alpha_{ik} + \beta_{jk}) \rho_{ik}]}{1 + \exp[\sum_{l=1}^K (\mu_k + \alpha_{ik} + \beta_{jk}) \rho_{ik}]} < \\ & (1 - \gamma) \sum_{j: \kappa_{jk}=1} (Y_{ij} - 0.5 * \rho_{ik}) \quad \text{and} \quad \tilde{\kappa}_{jk} = 1 \\ 0 & \text{otherwise} \end{cases}$$

where γ represents the reduction in residual sums of squares.

After the biclustering process is done, the task of inferring interacting pairs is still a hard one. Several criteria to decide which pairs of protein do interact were studied. Two of them best segregate the pairs of proteins that interact from those which do not interact: (1) The proportion of common biclusters for pair (i, j) , given by $(\# \text{biclusters where genes } i_1 \text{ and } i_2 \text{ are both present}) / \text{Maximum}(\# \text{biclusters where genes } i_1 \text{ is present, } \# \text{biclusters where genes } i_2 \text{ is present})$; and (2) The total number of bicluster columns for pair (i, j) , given by the sum of all unique columns of biclusters in which genes i_1 and i_2 both appear. Combining those two criteria leads to a better segregation; we adopted the rule that a pair of proteins interacts if at least one those values is above a given threshold.

4. Results

The Lbic model was compared to different models; the details of the experiments are given on Section 5.

We first compared the performances of the Lbic and the Plaid models when applied to phylogenetic data; the results are shown in Figure 1. It can be seen that the Lbic model achieves a much higher PPV score ($\text{PPV} = \text{TP} / (\text{TP} + \text{FP})$) throughout the entire curve, particularly among the top scoring pairs (lower sensitivity).

Because of the non-binary nature of the microarray expression data, the Lbic model cannot be applied to it. Nonetheless, since phylogenetic and

expression data contain different kinds of information, the Lbic model predictions can be improved by a combination with other prediction method based on expression data.

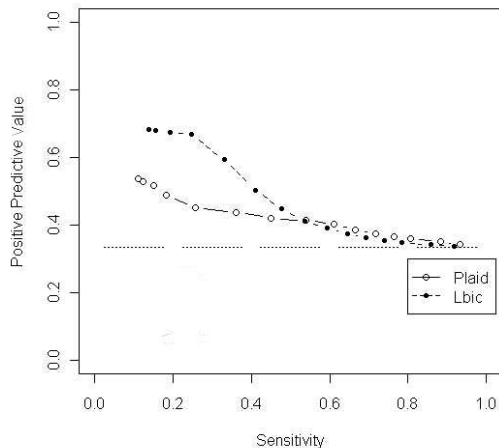


Figure 1. Lbic and Plaid model to phylogenetic data

While in certain cases the integration of several predictions based on different types of biological data yields worse results (probably due to the introduction of much noise) [10], we did a combination of the predictions made by applying the Lbic model to the phylogenetic data and by using the Plaid model to analyze the expression data, with changes in the PPV values. Figure 2 shows that the Plaid model applied to expression data is not as good as the Lbic model applied to the phylogenetic data.

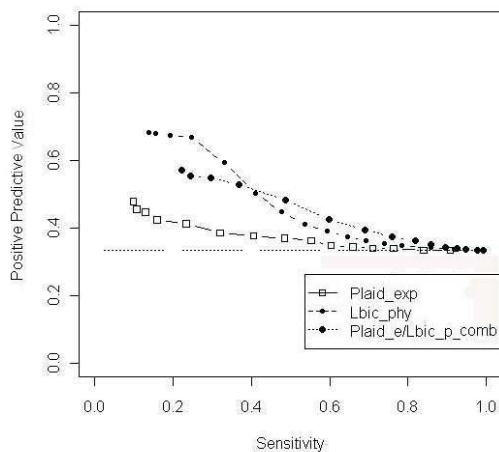


Figure 2. Lbic model to phylogenetic data and Plaid model to expression data and a combination of the two

The combined curve seems to have a little decrease of PPV value for small sensitivity, but it has a little increase for sensitivity around 0.45 to 0.65.

We also compared the performance of the combined predictions with those of the KCCA method proposed in Yamanishi et al. [20]. The data used for the KCCA method was the integrated one from Phylogenetic data and Expression data. The performance of supervised methods is highly dependent on the proportion of known information used in the training phase. Figures 3 shows the different results obtained by the KCCA method when trained using 10%, 25%, 50%, and 90% of the network as known data during the training phase. While the performances are obviously improved with a bigger percentage of knowledge, the scenario becomes more and more unrealistic as those proportions grow, since the actual knowledge of interaction networks is rather limited. We present that chart here in order to set the stage for a comparison between the performances of the KCCA supervised method with that of the combination Lbic and Plaid model, which are unsupervised.

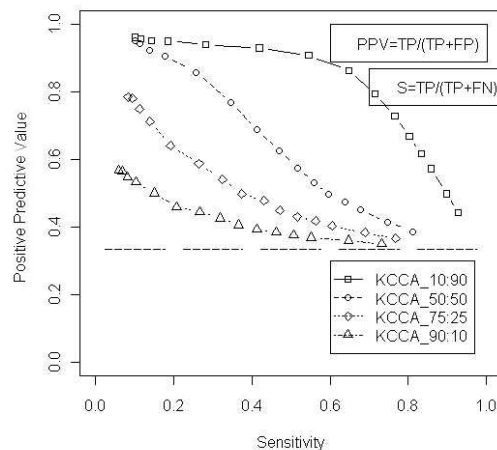


Figure 3. Phylogenetic data and expression data integrated from KCCA method

Figure 4 presents the comparison of the performances of the Lbic model with the KCCA method when trained with 10%, 25%, and 50% of the protein network, both using only phylogenetic data. We clearly see that the percentage of the previous knowledge is important to improve the performance of the KCCA method. The Lbic model is able to attain performances better than those three scenarios in the entire top half of the predictions. The phylogenetic data does not seem to be very informative for the KCCA method.

Figure 5 shows the prediction performance of the combination of (unsupervised) methods Lbic, using

phylogenetic data, and Plaid model, using expression data, in comparison with the performance of (supervised) method KCCA when trained with 25% and 10% of the protein network, using the integrated phylogenetic and expression data. We verify that while for our combined predictions the phylogenetic data is the principal dataset, for the KCCA method the expression data is the more relevant one. We observe that this combination has a performance very similar to the supervised method KCCA when trained with up to 25% of the gold standard network.

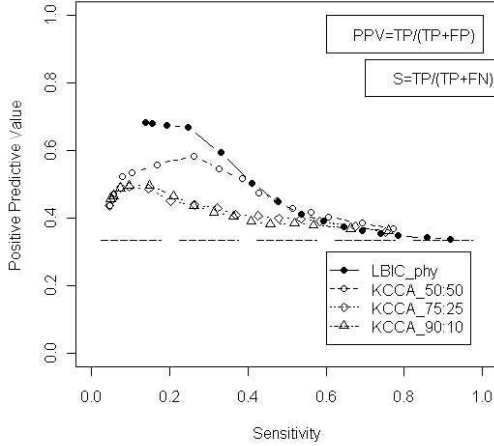


Figure 4. Lbic model and KCCA method to phylogenetic data

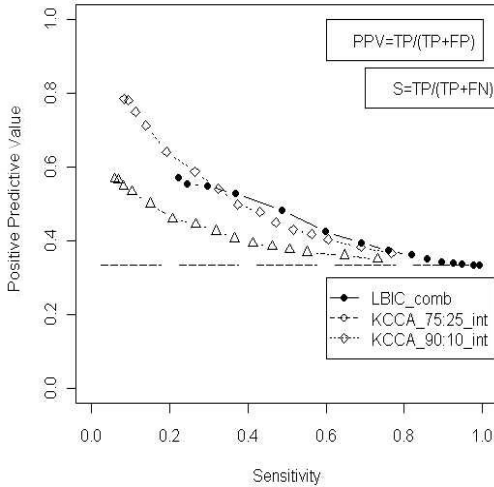


Figure 5. Phylogenetic data and expression data combined from Lbic and Plaid model, and phylogenetic data and expression data integrated from the KCCA method

In Tuner et al. [19] a partially supervised approach is introduced for the plaid model; we are looking in that direction to improve the Lbic model and the combined results.

5. Experiments Design

The Lbic and Plaid models were applied to *Phylogenetic profiles* and the Plaid model was applied to *Expression data* from 769 genes of the *Saccharomyces cerevisiae*. The data set used in this experiment is the same used in the experiments of Yamanishi et al. [20].

Expression Data correspond to results from 77 experiments in Spellman et al. [18] and 80 experiments in Eisen et al. [4]. The recorded information, Y_{ij} , represents the expression of the gene i under experiment j , where each experiment is seen as a condition.

Phylogenetic Profiles were built from the ortholog clusters in the KEGG database, which contains information about sets of orthologous genes in 145 organisms, where each organism is seen as a condition. Their recorded information, Y_{ij} , represents the presence or absence of an orthologous gene i under organism j .

The *Gold Protein Network Data* corresponds to a 769-protein network of *Saccharomyces cerevisiae* taken from the KEGG/PATHWAY database in Kanehisa et al. [8]. The recorded information, Y_{ij} , are elements from an adjacency matrix, which represents a graph with protein as vertices and interaction between proteins as edges. This data is considered as a reliable network.

For *Expression data* we run Turner's algorithm. The clustering process was applied to both rows and columns, and the algorithm was fitted to each layer using the parameters in Table 1.

Table 1: Parameters of Algorithm Turner

Row release criterion	NULL
Column release criterion	NULL
Number of permuted layers to use in permutation test	2
Number of times to back fit after each layer	4
Number of iterations to find a layer	6
Maximum number of layers to include in the model	15
Background	TRUE

For *Phylogenetic data* we run algorithms Lbic and Turner. For algorithm Lbic we considered that the clustering process also was applied to both rows and columns and the algorithm was fitted to each layer using the parameters in Table 2. The parameters of the model were estimated using Newton Raphson's method.

Table 2: Parameters of Lbic algorithm

Row release criterion	0.8
Column release criterion	0.8
Number of permuted layers to use in permutation test	2
Number of times to back fit after each layer	4
Number of iterations to find a layer	3
Maximum number of layers to include in the model	10

For algorithm Turner, we consider the same model and parameters used for the expression data. The parameters chosen for those data sets were chosen in order to better segregate pairs that interact from those who do not interact.

In order to verify if the Lbic model fit well the phylogenetic data we compare it with the Plaid model. This comparison was made through the graphic of sensitivity, S ($S = TP/(TP+FN)$), versus positive predictive value, PPV ($PPV = TP/(TP+FP)$).

Those two models do not necessarily cover all pairs of genes associated to the gold standard protein network (GSPN), in the sense that there are pair of genes that do not belong to any biclusters. Hence we made the graphic only for those genes that belong to at least one bicluster. For phylogenetic data we get around 300 pairs of genes that do interact in the GSPN while using expression data we got around 600 pairs. Because we have a great difference between the number of pairs that do interact and the number of pair that do not interact in the GSPN, we randomly choose pairs from those that do not interact in the GSPN, in proportion 1:2, that is, the number of pairs that do not interact in the GSPN is twice the number of pairs that do interact. Hence, for phylogenetic data we got around 900 pairs, and for expression data we got around 1800 pairs. We repeat that process a number of times, and then we plot the average values of PPV and sensitivity over the same thresholds in all runs. The thresholds were chosen taking into account the ordered values of the chosen criterion. We also consider the number of pairs of the network that the model gets to infer, since the

Lbic and the Plaid models do not cover the same number of pairs of genes. Hence the thresholds are proportions of the number of pairs to be inferred. For example, the first threshold is set at the 3,5% top scored values, the second is set at the 4% top scored values and so on for 5%, 6,5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, and 70%.

In order to get information from both datasets, expression and phylogenetic, we combine their results in a naïve way, saying that a pair interacts if at least one dataset says so. Hence the combined sensitivity (S_c) and specificity (E_c) were found as:

$$S_c = S_{phy} + S_{exp} - S_{phy} * S_{exp}$$

and

$$E_c = E_{phy} * E_{exp},$$

where S_{phy} is the sensitivity from phylogenetic data, S_{exp} is the sensitivity from expression data, E_{phy} is the specificity from phylogenetic data, and E_{exp} is the specificity from expression data. Therefore the PPV_c is given by:

$$PPV_c = S_c * (1/3) / [S_c * (1/3) + (1 - E_c) * (2/3)],$$

where 1/3 is the probability that one pair interacts, since we selected pairs from those that do not interact in the GSPN in proportion 1:2.

For the KCCA method we used all parameters presented by Yamanishi et al. [20], but we used different proportions of GSPN for testing/training the dataset. Besides the proportion used in their paper (10:90), the proportions 50:50, 75:25 and 90:10 for testing/training were also presented.

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8. References

- [1] Aytuna, A.S., Gursoy, A., and Keskin, O. Prediction of protein-protein interactions by combining structure and sequence conservation in protein interfaces. *Bioinformatics*. 2005, 21(12)
- [2] Brown K.R., Jurisica I. Unequal evolutionary conservation of human protein interactions in interologous networks. 2007, 8:R95.
- [3] Christensen, R. Log-Linear Models and Logistic Regression. *Springer-Verlag*, second edition, New York. 1997.
- [4] Eisen, M.B., Spellman, P.T., Patrick, O.B. and Botstein, D. Cluster analysis and display of genome-wide

- expression patterns. *Proc. Natl. Acad. Sci.* 1998, 95:14863-14868.
- [5] Espadaler, J., Romero-Isart, O., Jackson, R.M. and Oliva, B. Prediction of protein-protein interactions by combining structure and sequence conservation in protein interfaces. *Bioinformatics.* 2005, 21(16).
- [6] Guimarães, K.S., Jothi, R., Zotenko, E. and Przytycka T.M. Predicting domain-domain interactions using a parsimony approach. *Genome Biology.* 2006, 7: R104.
- [7] Hosmer, D.W. and Lemeshow, S. Applied Logistic Regression. *John Wiley & Sons, Inc.*, second edition, New York. 2000.
- [8] Kanehisa, M., Goto, S., Kawashima, S., Okuno, Y., Hattori, M. The KEGG resources for deciphering the genome. *Nucleic Acids Res.* 2004, 32:D277-D280.
- [9] Lazzeroni, L. and Owen, A. Plaid Models for Gene Expression Data. *Statistica Sinica.* 2002, 12:61-86.
- [10] Lee, H., Deng, M., Sun, F. and Chen, T. An integrated approach to the prediction of domain-domain interactions. *BMC Bioinformatics.* 2006, 7:269.
- [11] Madeira, S.C. and Oliveira, A.L. Biclustering algorithms for biological data analysis: A survey. *IEEE Transactions on Computational Biology and Bioinformatics.* 2004, 1:24-45.
- [12] Myers, C.L. and Troyanskaya, O.G. Context-sensitive data integration and prediction of biological networks. *Bioinformatics.* 2007, 23(17):2322-2330.
- [13] Pellegrini, M., Marcotte, E.M., Thompson, M.J., Eisenberg, D. and Yeates, T.O. Assigning protein function by comparative genome analysis: Protein phylogenetic profiles. *Proc. Natl. Acad. Sci.* 1999, 96:4285-4288.
- [14] Ralambondrainy, H. A conceptual version of the K-means algorithm. *Pattern Recognition Letters.* 1995, 16:1147-1157.
- [15] Reiss, D. J., Baliga, N.S. and Bonneau, R. Integrated biclustering of heterogeneous genome-wide datasets for the inference of global regulatory networks. *BMC Bioinformatics.* 2006, 7:280/1471-2105.
- [16] Riley, R., Lee, C., Sabatti, C. and Eisenberg, D. Inferring protein domain interactions from databases of interacting proteins. *Genome Biology.* 2005, 6:R89.
- [17] Searle, S.R. Linear Models. *John Wiley & Sons, Inc.* New York. 2000.
- [18] Spellman, P.T., Sherlock, G., Zhang, M.Q., Iyer, V.R., Andres, K., Eisen, M.B., Brown, O.O., Botstein, D. and Futcher, B. Comprehensive identification of cell cycle-regulated genes of the yeast *Saccharomyces cerevisiae* by microarray hybridization. *Mol. Biol. Cell.* 1998, 9:3273-3297.
- [19] Tuner, H.L., Bailey, T.C., Krzanowski, W.J. and Hemingway, C.A. Biclustering models for structured microarray data, *IEEE Transactions on Computational Biology and Bioinformatics.* 2005, 2:316-329.
- [20] Yamanishi, Y., Vert, J.-P., and Kanehisa, M. Protein network inference from multiple genomic data: a supervised approach. *Bioinformatics.* 2004, 20:i363-i370.